

Board gender diversity: the hidden impact of female directors on environmental decoupling

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Abstract

Purpose – The purpose of this paper is to explore the influence of board gender diversity on environmental decoupling – including both greenwashing and brownwashing – viewed as a critical barrier to corporate accountability and sustainable development.

Design/methodology/approach – This study analyzes a sample of US listed companies from 2011 to 2021, employing various methodological approaches and robustness tests to ensure the accuracy and reliability of our results.

Findings – The results highlight that female directors can increase greenwashing and reduce brownwashing. This effect is amplified in environmentally sensitive industries where the pressure for sustainable development is higher. However, in firms with low levels of environmental, social and governance (ESG) controversies, board gender diversity may contribute to greenwashing.

Practical implications – The findings provide insights for firms to understand what effect board gender diversity has on substantive environmental actions. For policymakers, results suggest that gender measures must be complemented by robust accountability mechanisms to effectively mitigate the risks of environmental decoupling and support sustainable development.

Social implications – This study underscores the potential unintended consequences of female board representation and highlights the need to address decoupling to protect society from misleading information and thereby ensure that corporate transparency contributes to genuine sustainable development.

Originality/value – Moving beyond a generic ESG focus, the study provides a more nuanced understanding of how board gender diversity affects environmental decoupling, positioning it as a critical factor that determines corporate transparency and its ultimate contribution to sustainable development.

Keywords Board gender diversity, Environmental decoupling, Sustainable development, Greenwashing, Brownwashing

Paper type Research paper

1. Introduction

The integration of environmental, social and governance (ESG) activities into business strategies has become a major challenge for companies that carry out both internal actions, such as implementing ESG initiatives (performance) and external actions related to ESG



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disclosures (reporting) (Velte, 2023). However, firms may adopt decoupling practices when these two dimensions are not aligned (Gull *et al.*, 2023a). This ESG gap – defined as the degree of inconsistency between ESG reporting and initiatives (Hawn and Ioannou, 2016; Tashman *et al.*, 2019; García-Sánchez *et al.*, 2022) – may manifest itself in two ways. A positive gap (greenwashing) occurs when a firm’s ESG reporting exceeds its actual practices, whereas a negative gap (brownwashing) arises when a firm underreports its ESG practices (Kim and Lyon, 2015). Our paper uses these terms to describe types of decoupling, and emphasizes that they do not necessarily imply intentional or deliberate misreporting but that they may instead represent unintended consequences arising from the relative balance between external reporting and internal environmental initiatives. From a sustainable development perspective, addressing this balance is crucial, as ESG strategies are key mechanisms through which organizations discharge their accountability and convey their contribution to long-term sustainability objectives.

ESG decoupling has emerged as a significant concern for policymakers (European Commission, 2022 [1]; European Commission, 2023 [2]; ESMA, 2023; IFRS Foundation, 2023a, 2023b; IOSCO, 2023) and professional organizations (Deloitte, 2022; Ernst and Young, 2022; PwC, 2022) due to its significant consequences for both society and firms. Recent literature has emphasized this decoupling as a critical issue (Jauernig and Valentinov, 2019; Talpur *et al.*, 2024; Palea *et al.*, 2025) since, irrespective of the direction of the ESG gap, it is likely to undermine firms’ credibility regarding sustainable development by weakening the trustworthiness of ESG initiatives (Eliwa *et al.*, 2023). This problem could harm companies by worsening access to finance (García-Sánchez *et al.*, 2021), damaging corporate reputation (Miras-Rodríguez *et al.*, 2020) and firm value (Hawn and Ioannou, 2016), which are outcomes that are particularly relevant for policymakers, regulators and stakeholders, who rely on credible ESG reporting to guide regulation, monitor corporate behavior and ensure transparent accountability in decision-making processes.

In light of these concerns, understanding the factors that can influence the ESG gap represents a critical challenge in the literature. The empirical evidence reported in previous research has thus far suggested that the ESG gap may be determined by external pressures from governments (Marquis and Qian, 2014; Luo *et al.*, 2017), the societal and institutional environment (Cho *et al.*, 2015; Tashman *et al.*, 2019), industry (Kim *et al.*, 2017), regulations (Kim and Lyon, 2015; Aboud *et al.*, 2024) or analysts (García-Sánchez *et al.*, 2022). Other studies have documented that the ESG gap may be influenced by firm characteristics, such as firm size (Maas and Liket, 2011), performance (Kim and Lyon, 2015) and ownership structure (Liu *et al.*, 2023). Among the internal factors involved, corporate governance – and in particular board composition – has garnered increasing attention. The thus far limited research has examined whether the ESG gap may be determined by CEOs (Shahab *et al.*, 2022; Gull *et al.*, 2023c), governance committees (Gull *et al.*, 2023b), sustainability committees (Gull *et al.*, 2024) or other board characteristics, such as directors’ networks (Zhao *et al.*, 2022) and directors’ independence (Yu *et al.*, 2020).

One area that remains underexplored is the analysis of how ESG decoupling may be determined by board gender diversity, and it is an issue found on the agenda of the main international strategies (i.e. in the European Union and the USA [3]), as well as in corporate governance codes throughout the world. The scarce empirical evidence regarding what impact female board members might have on ESG decoupling remains mixed, as studies have found that female directors may decrease the ESG gap (Eliwa *et al.*, 2023), increase the ESG gap (Shahab *et al.*, 2022) or have a nonsignificant effect (Gull *et al.*, 2023b). However, these studies vary in terms of samples and periods and focus on aggregated ESG measures rather than on individual dimensions. Moreover, they do not tend to distinguish between

greenwashing and brownwashing and they fail to account for relevant contextual factors that may moderate the influence of female directors. There is therefore still insufficient critical mass to provide robust evidence on this topic, which underlines the need for further research to adopt effective measures (Bannò *et al.*, 2023).

Our paper addresses these limitations in three ways. First, we analyze the unintended consequences of board gender diversity on specific environmental decoupling. In an effort to provide more accurate insights, it is crucial to break down ESG components (Aguilera *et al.*, 2021; Dinh *et al.*, 2023), especially as environmental reporting lies at the heart of recent international reforms (i.e. Environmental Sustainability Reporting Standards and IFRS Sustainability Standards). Second, we disentangle the concept of decoupling, which is key to gaining a better picture of the effects of board gender diversity (Gull *et al.*, 2023c; Gull *et al.*, 2024). We thus examine its influence on the different signs of the environmental gap, since motivations for greenwashing and brownwashing differ. While companies may engage in greenwashing to enhance corporate reputation, meet stakeholder expectations, or comply with social and regulatory norms (Delmas and Burbano, 2011; Kim and Lyon, 2015), brownwashing can be explained by a desire to avoid disclosing costly practices, as well as a fear of criticism from external stakeholders (Kim and Lyon, 2015; Montgomery *et al.*, 2024). Third, our paper overcomes previous limitations by exploring how new contextual factors may moderate the influence of female directors on the environmental gap.

Specifically, our study presents a twofold objective to contribute to this line of research. First, our paper performs an in-depth analysis to explore whether female directors influence environmental decoupling –focusing separately on greenwashing and brownwashing. Second, our paper investigates how external factors that may increase pressures on ESG strategies can moderate the role of female directors in environmental decoupling. In this regard, we consider both the environmental sensitivity of the industry in which a firm operates and its level of ESG controversies. One methodological novelty of this paper is that we consider how the relation between board gender diversity and environmental decoupling is simultaneously moderated by the industry and by a firm's level of ESG controversies, thereby enabling us to obtain more accurate information regarding the moderation analyses.

Our results highlight that there is a relation between board gender diversity and environmental decoupling, with female directors specifically tending to reduce the environmental gap. Nonetheless, the effect is different when considering the type of decoupling in which companies incur. In particular, board gender diversity appears to increase greenwashing while reducing brownwashing. In addition, our results reveal that the influence of female directors on the environmental gap is moderated by external factors. Regarding general industry pressures, results reveal that female directors have a greater effect on both the increase in greenwashing and the decrease in brownwashing in environmentally sensitive industries. However, the moderating effect of firm-level pressures is different, as female board directors exert a greater influence on greenwashing practices in firms in which there is a reduced level of environmental pressure due to there being a low level of ESG controversies. These findings make several significant contributions to the literature.

First, we expand the empirical evidence concerning the influence of corporate governance mechanisms on ESG decoupling (Shahab *et al.*, 2022; Zhao *et al.*, 2022; Gull *et al.*, 2023b), and which proves relevant *vis-à-vis* adopting effective measures. Whereas previous studies have identified external factors – such as policymaker and stakeholder influence – that impact ESG decoupling, the role of boards has been less explored, even though they are at the center of public and academic debate, given their crucial role as governing mechanisms within companies in terms of developing ESG strategies (Velte, 2023). Our findings add

evidence to the discourse on sustainable development by showing that boards are a key channel through which both environmental greenwashing and brownwashing practices are determined. Second, our evidence contributes to the scarce literature regarding board gender diversity and ESG decoupling. As mentioned, the empirical evidence to date remains inconclusive, which is likely explained by female directors possibly having a different impact on ESG decoupling depending on various circumstances. Our study shows that women directors reduce the environmental gap in general terms. However, in-depth analysis reveals that board gender diversity actually increases greenwashing and decreases brownwashing, which may have important ethical implications, and which opens the door to new avenues for research regarding the varying effect that female directors have on environmental reporting and initiatives. Our study thus reconciles previous research and emphasizes the need for further detailed investigation. Third, our findings advocate the idea that a more nuanced approach is needed to truly comprehend what impact female directors have on ESG strategies (García-Meca and Martínez-Ferrero, 2025). In this vein, our study offers a pioneering view and explores the nexus between female directors and environmental decoupling, considering two contextual factors that can determine external pressures. Our findings contribute to the literature by shedding light on the contrasting effects of these factors, illustrating that women directors react differently to general industry pressures compared to specific firm-level pressures related to ESG controversies. Ultimately, focusing on environmental decoupling as a governance challenge allows for a deeper understanding of how corporate practices and policy solutions can address sustainability hurdles, thereby moving beyond a purely financial view of ESG toward one centered on genuine transparency.

2. Theoretical framework and hypothesis development

Decoupling occurs when corporate sustainability reporting does not fully correspond to firms' actual environmental initiatives – a phenomenon often referred to as greenwashing or in some cases brownwashing. This decoupling has been measured using traditional ESG metrics, including ESG scores and reporting indicators (Hawn and Ioannou, 2016; García-Sánchez *et al.*, 2022), as well as more recent textual analysis techniques such as natural language processing and machine learning applied to corporate reports (Conrad and Holtbrügge, 2021; Gorovaia and Makrominas, 2025).

Researchers have increasingly studied the influence of female directors on sustainability strategies, and suggested that board gender diversity may have an effect on ESG decoupling (Eliwa *et al.*, 2023). While no single theory can provide a fully comprehensive framework to explain the relation between female directors and sustainability strategies, gender socialization theory (Stockard, 1999) has been extensively used in this field of research (Carvajal *et al.*, 2022).

Gender socialization theory (Stockard, 1999) emphasizes the differences in behavior between female and male directors based on their distinct psychological characteristics. Previous literature widely agrees that these differences can explain how female board members may have a significant impact on corporate environmental strategies (Arayssi *et al.*, 2019; Khatri, 2023). In this regard, women tend to have more communal, supportive and empathic traits (Eagly *et al.*, 2003) and are generally more sensitive to sustainable development (Harjoto and Rossi, 2019). They are also likely to display a morality that is more focused on long-term caring, and female directors can pay more attention to environmental and social issues (Alkhawaja *et al.*, 2023). From an accountability perspective, this heightened sensitivity suggests that women directors are more likely to act as a mechanism to discharge the firm's responsibility toward society (Al-Shaar and Zaman, 2016). Women directors are thus

likely to bring social and human capital to boards that is more focused on sustainability matters (Ramon-Llorens *et al.*, 2021), thereby increasing awareness toward enhancing corporate environmental responsibility (Kyaw *et al.*, 2022).

However, the empirical evidence remains inconclusive. Emerging studies have found that female directors may both reduce ESG decoupling (Eliwa *et al.*, 2023) and increase ESG decoupling (Shahab *et al.*, 2022), while other papers have failed to find any significant relation between board gender diversity and ESG decoupling (Gull *et al.*, 2023b), or have found a negative effect of female directors on ESG decoupling only when these women are nonexecutive (Gull *et al.*, 2023a). In the field of environmental decoupling, recent studies (Gull *et al.*, 2023c; Gull *et al.*, 2024) emphasize the need to consider the sign of decoupling, and although these papers focus on CEOs and sustainability committees, the results show that female directors – considered as a control variable – may have different effects on brownwashing and greenwashing.

These contradictory findings demand more in-depth theoretical insights to understand how the effect of women in the boardroom maps onto environmental decoupling. In this regard, we argue that female directors – without necessarily intending to engage in unethical behavior – might influence environmental decoupling by having a more immediate and effective influence on environmental external actions (reporting) than on environmental internal actions (initiatives).

On the one hand, environmental initiatives require complex decision-making processes and huge investments (Hussain *et al.*, 2018), including regulatory and technological aspects (Haque, 2017) and high uncertainty regarding future returns (Khoo *et al.*, 2022). While environmental initiatives involve higher costs and organizational burdens (Miska *et al.*, 2018; Liang *et al.*, 2022), environmental reporting is generally easier to implement and may constitute a short cut to addressing societal and stakeholder concerns (Matuszak *et al.*, 2019). On the other hand, the reduced level of female directors' actual power may prove to be a handicap when seeking to implement complex environmental strategies (Saggese *et al.*, 2021). As a minority group, women may find it difficult to become effective actors in the most challenging capital-allocation decisions regarding environmental strategies (Cook and Glass, 2015), which are typically linked to internal environmental initiatives. What is more, female directors might be employed as a signal to markets (Fasan and Mio, 2017) and may have only a limited impact on operational environmental initiatives (Yarram and Adapa, 2021), with it being easier for these women to push boards toward reporting.

This asymmetric influence directly impacts the decoupling gap. If female director emphasis on corporate environmental strategies leads boards to more rapidly increase environmental reporting than to actually engage in costly environmental initiatives (Eliwa *et al.*, 2023) –and even if it is not intentionally unethical behavior (Bravo-Urquiza and Reguera-Alvarado, 2024)– the alignment between reporting and initiatives necessarily shifts, which can affect the extent and type of environmental decoupling. Consequently, this effect may vary depending on the firm's underlying communication strategy. In this regard, greater emphasis on environmental reporting over initiatives would increase decoupling in firms engaged in greenwashing (as the gap between reporting and actual initiatives widens) and would decrease it in firms engaged in brownwashing (as the gap narrows).

Taking into consideration the arguments above, the following hypothesis is formulated:

- H1.* Board gender diversity influences environmental decoupling by shaping the balance between environmental reporting and environmental initiatives, thereby affecting the integrity of corporate accountability.

Given that directors operate within a specific environment that shapes their decision-making, our paper aligns with the recent literature highlighting that the role of boards in ESG outcomes is expected to vary significantly depending on contextual factors (Endrikat *et al.*, 2021). In this regard, the influence of female board members on environmental decoupling might be conditioned by various external pressures (Talpur *et al.*, 2024), which may include both general industry pressures and firm-level pressures derived from ESG controversies reported in the media about the firm. In our study, these external pressures are considered as moderating factors of the influence of board gender diversity on environmental decoupling.

2.1 *The moderating role of environmentally sensitive industries*

Firms operating in environmentally sensitive industries face increased pressure regarding sustainability outcomes and must meet greater expectations from various stakeholders concerning ESG aspects (Böhling *et al.*, 2019). Stakeholders in such industries can exert significant pressure and substantial scrutiny concerning ESG issues (Zaiane and Ellouze, 2023), and firms might be more aware of the need to build trust through their environmental strategies (Emma and Jennifer, 2021). In this sense, environmental reporting has become vital in these environmentally sensitive industries, thereby increasing the pressure on firms to persuade stakeholders that the company is carrying out responsible environmental activities (Hawn and Ioannou, 2016). In this regard, firms are likely to strengthen their environmental reporting to respond to increased regulatory pressures (Meng *et al.*, 2019) and so avoid losing competitive advantage (Eliwa *et al.*, 2023). In many cases, these firms can overreport environmental issues because stakeholders have a low degree of proximity to firms (Ruiz-Blanco *et al.*, 2022). Such companies might even perceive that it is difficult for stakeholders to assess environmental strategies (Rodrigue *et al.*, 2013).

Building on the arguments from H1, female directors could exert a more immediate effect on environmental reporting rather than on environmental actions. In environmentally sensitive industries – where external pressures from stakeholders, regulators and the public are higher – the influence of female directors is likely to be more pronounced, as heightened board awareness can translate more effectively into observable actions. Taken together, these mechanisms suggest that the effect of board gender diversity on environmental decoupling is stronger in environmentally sensitive industries, thus leading to the following hypothesis:

- H2. The influence of board gender diversity on environmental decoupling is stronger in environmentally sensitive industries.

2.2 *The moderating role of firm-level environmental, social and governance controversies*

Firms with high levels of ESG controversies may be subject to significant stakeholder distrust, with their sustainability strategies usually coming under intense scrutiny (Galbreath, 2013), which thus increases the pressure related to environmental decision-making. Furthermore, media coverage of irresponsible behavior could be viewed as a signal of poor corporate governance mechanisms, and in particular of inadequate board supervision (Van Scotter and Roglio, 2020). Nevertheless, how company boards respond to these pressures remains unclear and the literature on this topic is very scarce (Issa, 2023). Firms under greater pressure due to an unwanted reputation that is echoed through the media because of the level of ESG controversies can often attract the attention of regulatory agencies and nongovernmental organizations who may impose stricter oversight. These firms are aware of the need to restore their image to gain legitimacy and they send out positive signals to the financial market in an effort to ensure that their market value does not suffer (Shakil, 2021).

However, firms with a higher level of ESG controversies may be prone to maximize short-term value creation and prioritize near-term outcomes (Shi *et al.*, 2020). In this vein, firms are more involved in external ESG activities than internal ones, thereby proving their desire to seek visibility (Al-Shammari *et al.*, 2019). ESG controversies thus often give rise to pressures on companies and compel them to address stakeholder concerns by increasing environmental reporting as a first response – aiming to mitigate the negative repercussions – since ESG disclosure can be used to boost and protect legitimacy (Cho *et al.*, 2015).

In firms that face greater ESG controversies, the heightened external scrutiny and stakeholder attention amplify the influence of female directors, as their awareness can more quickly translate into observable actions, particularly in terms of advancing environmental reporting relative to internal initiatives. Taken together, these mechanisms suggest that the effect of board gender diversity on environmental decoupling is stronger in firms with greater ESG controversies, which directly leads to the following hypothesis:

- H3.* The influence of board gender diversity on environmental decoupling is stronger in firms with greater ESG controversies.

The conceptual framework is shown in Figure 1.

3. Methodology

3.1 Data and sample

Our sample is composed of firms listed on the S&P 500 for the period 2011–2021. These companies were selected because they represent the largest and most influential publicly traded companies of the US market. They also display high visibility, significant stakeholder engagement and generally more advanced ESG disclosure practices. The selected period (2011–2021) ensures that the results are not biased by the aftermath of the 2008 financial crisis and allows for the observation of stable trends in the performance of S&P 500 companies prior to the proposal of recent mandatory ESG disclosure requirements. Our initial sample consists of 3,583 observations, which is the result of merging the required databases. This number is small due to missing values – particularly those related to the variable on controversies. In addition, the time lag required to compute the variable used to quantify environmental decoupling further reduces the sample size. Finally, we removed outliers from two control variables: firm leverage and board tenure. As a result, our analysis is based on an unbalanced panel of 2,002 observations, representing 347 unique S&P 500 firms.

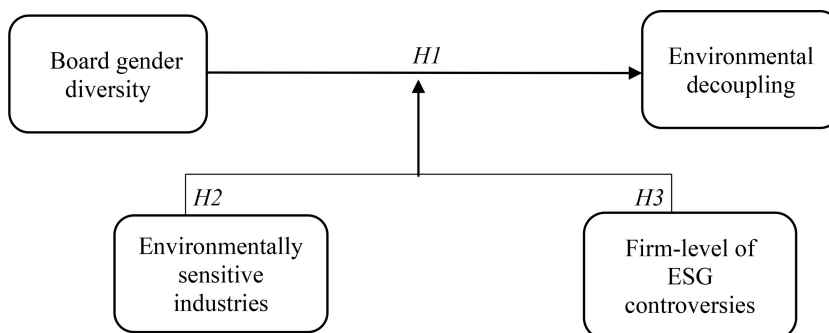


Figure 1. Conceptual framework

To ensure the comparability and reliability of our findings, data for the design of the main variables of our analysis are collected from widely recognized sources in the previous literature. Information concerning environmental decoupling is compiled from the LSEG Workspace database, financial data are obtained from Compustat and information about board gender diversity and other board of directors' characteristics is extracted from BoardEx. These databases were selected due to their reliability and wide recognition in the literature as well as their provision of audited data, which ensures comparability and replicability.

3.2 Variables design

3.2.1 Dependent variable: environmental gap. To measure environmental decoupling – which reflects the inconsistency between environmental reporting and initiatives – the variable environmental gap (Env_GAP) is employed. In line with prior studies (Hawn and Ioannou, 2016; García-Sánchez et al., 2022; Gull et al., 2024), this variable captures the gap between the scores of 22 items related to external actions (reporting) – measured one year ahead – and 21 items related to internal actions (initiatives), obtained from LSEG Workspace database, since reporting typically follows internal implementation [4]. First, we add up each of the reporting items and each of the items related to initiatives to create two intermediate variables: the environmental reporting variable, and the environmental initiative variable. Second, each of these variables is normalized [5] using min-max scaling, taking into account the minimum and maximum values within the sample. Finally, Env_GAP is calculated as the difference between the scores for current environmental reporting and the scores for the lagged environmental initiatives. As a result, Env_GAP can range between -1 and + 1, where higher scores indicate greater misalignment between a firm's environmental reporting and its initiatives. Two additional dependent variables are considered to separately test the influence of board gender diversity on both the positive environmental gap (GREENWASH) or greenwashing, and the negative environmental gap (BROWNWASH) or brownwashing.

3.2.2 Explanatory variables and control variables. Two alternative explanatory variables are mainly employed to capture board gender diversity. First, the percentage of female board members (*Prop_wom*) is considered. Second, the Blau index of heterogeneity (*Blau*) is employed. It is calculated as $D = 1 - \sum p_i^2$, where *p* represents the proportion of individuals belonging to each category (fraction of female and male directors), and *i* is the number of categories (two in our case) (Blau, 1977).

Moreover, a set of control variables is also included in the empirical analysis. We control for the effect of several board attributes and firm-specific characteristics that are calculated for each year of interest in our analysis. Regarding the board of directors, the following control variables are considered: the existence of a CSR or sustainability committee (*Sust_com*), which is a dummy variable that equals 1 if the company presents this committee, and 0 otherwise; board independence (*BIndep*), computed as the proportion of independent directors on the board; board size (*BSize*), calculated as the total number of directors on the board; ESG experience of the board (*BESG_exp*), which is defined as the proportion of directors who are also on the boards of other firms that operate in ESG sensitive industries. As regards firm-specific characteristics, the following variables are considered: firm size (*FSize*), defined as the logarithm of total assets; firm performance (*FPerf*), calculated as the return on assets; firm leverage (*FLev*), computed as the ratio of total debt to total assets; firm-level of ESG controversies (*ESG_controv* [6]), which is defined as the level of ESG controversies of a firm disclosed in the media, measured on a scale from 0 to 100, where 0 represents maximum media pressure due to a high level of ESG controversies; belonging to an environmentally sensitive industry (*Env_industry* [7]), which is a dummy variable that

takes the value 1 if the company belongs to an environmentally sensitive industry (chemical, paper, metals, petroleum, mining and extractive and utility industries – chemicals, metals, mining, paper, petroleum or utilities industries as being environmentally sensitive) following the SIC codes classification (Fernandez-Feijoo *et al.*, 2014), and 0 otherwise.

Table 1 shows a summary of all the variables included in our analysis, together with their definitions.

3.3 Empirical analysis

Our empirical analysis tests the effect of female directors on environmental decoupling. The general model is presented in equation (1), where β_0 is the intercept, β is the coefficient of each independent variable, sub index i identifies the firm, subindex t identifies time and ε_{it} is the error term:

$$\begin{aligned} Env_GAP_{it+1} = & \beta_0 + \beta_1 Prop_wom(Blau)_{it} + \beta_2 Sust_com_{it} + \beta_3 BIndep_{it} + \beta_4 BSize_{it} \\ & + \beta_5 ESG_exp_{it} + \beta_6 FSize_{it} + \beta_7 FPerf_{it} + \beta_8 FLev_{it} + \beta_9 Env_industry \\ & + \beta_{10} ESG_controv_i + \beta_{11} Year_t + \varepsilon_{it} \end{aligned} \quad (1)$$

In relation to board gender diversity, our two measures (Prop_wom and Blau) are alternatively used in all the models to guarantee that our findings are not biased by the design of our main explanatory variable. Our empirical approach consists of several stages to ensure the reliability of our findings and to address our research hypotheses.

First, our general model is applied by employing the generalized method of moments (GMM). To ensure the adequacy of this procedure, the Arellano-Bond test (AR1) in first differences and the Sargan test of overidentification are applied. GMM provides consistent

Table 1. Variables description

Name	Variable definition
Env_GAP	Difference in the scores of 22 items for external actions (reporting) and 21 items for internal actions obtained from LSEG Eikon
Prop_wom	Proportion of female board members
Blau	Blau index of heterogeneity
Sust_com	Dummy variable that equals 1 if the company presents this committee, and 0 otherwise
BIndep	Proportion of independent directors
BSize	Total number of directors on the board
BESG_exp	Proportion of directors with specific experience serving on boards of other firms operating in ESG sensitive industries
FSize	Logarithm of total assets
FPerf	Return on assets
FLev	Ratio of total debt to total assets
Env_industry	Dummy variable that takes the value 1 if the company belongs to an environmentally sensitive industry (following SIC codes classification)
ESG_controv	Level of ESG controversies of a firm, based on 23 ESG controversy topics ^a provided by LSEG Workspace database; ranges from 0 to 100, where 0 symbolizes that media pressure is maximum due to the high level of ESG controversies (LSEG Data and Analytics, 2024)

Note(s): ^aAppendix of the report Environmental, Social, and Governance scores from LSEG (LSEG Data and Analytics, 2024) – called Controversies Data Measures – relates each item considered to quantify this variable

estimators by using internal instruments, thus safeguarding reliable estimates in the presence of simultaneity, omitted variable bias or reverse causality – issues that OLS cannot adequately handle. This methodology is widely applied in corporate governance and sustainability research (García-Sánchez *et al.*, 2022; Gull *et al.*, 2024; Grau Grau *et al.*, 2025). In addition, to control endogeneity issues, three different estimation models are also performed: fixed effect linear regression for panel data using the lagged independent variable, two stage least squares (2SLS) and simultaneous equation approaches. Regarding the fixed-effects linear regression model, the main explanatory variable is lagged (García-Sánchez *et al.*, 2022) since, although current values of board gender diversity might be endogenous to our dependent variables, this problem is unlikely to appear for past values of board gender diversity. In relation to the 2SLS approach, we employ both the percentage of women on the nominating committee and board tenure as instrumental variables. The validity of these instruments is confirmed by the values obtained from the Sargan test – reported in the subsequent section. With reference to the simultaneous equations methodology (Young *et al.*, 2008), the first equation represents the general model proposed in our study, while the second equation seeks to elucidate the determinants of gender diversity.

To strengthen the causal interpretation of our findings, we complement our main analysis with a difference-in-differences (DiD) approach, exploiting the 2018 California board gender quota law as a quasi-natural experiment. This regulation mandated a minimum number of women on corporate boards, thus providing an exogenous shock to board gender composition in affected firms. By comparing Californian firms to those in other US states before and after the law, we aim to isolate the effect of increased female board representation on environmental decoupling. The treatment group consists of all California-based firms, while the control group includes never-treated firms located in other states. To ensure the robustness of our identification strategy, we assume an absorbing treatment and explicitly address the risk of selection endogeneity; we excluded from the sample any firm that relocated (either into or out of California) during the study period. This strict group definition ensures that the estimated effect can be credibly attributed to policy change.

In addition, the validity of the parallel trends assumption – which is key to the DiD design – is visually assessed by constructing an event study plot showing the estimated coefficients for both pre and posttreatment periods. This graphical representation is essential for validating the identification strategy and for analyzing the temporal dynamics of the policy's effect.

Second, a more specific analysis for each type of environmental gap – greenwashing and brownwashing – is carried out. This analysis provides a better understanding of the influence of female directors on environmental decoupling and helps to address our *H1*.

Third, moderation analyses are performed to test our *H2* and *H3* and to determine how external pressures can moderate the influence of female directors on environmental decoupling. Specifically, two moderation factors are considered: belonging to an environmentally sensitive industry, and firm-level ESG controversies.

Fourth, an additional analysis is performed to provide more details on how the relation between board gender diversity and environmental decoupling is moderated by the industry and the level of a firm's ESG controversies. To that end, a triple interaction between the variables board gender diversity, Env_industry and ESG_controv is carried out.

4. Results

Table 2 displays the main descriptive statistics. The mean value of the Env_GAP is -0.090 , highlighting that scores for reporting are, on average, lower than scores for initiatives, which

Table 2. Descriptive statistics

Variable	Mean	SD	Q1	Median	Q2	Min.	Max.
Env_GAP	−0.090	0.162	−0.196	−0.086	0.118	−0.698	0.541
Prop_wom	0.222	0.100	0.154	0.222	0.273	0	0.667
Blau	0.326	0.108	0.261	0.345	0.397	0	0.5
BIndep	0.855	0.083	0.818	0.889	0.909	0.4	1
BSize	10.984	2.190	10	11	12	5	30
BESG_Eexp	0.498	0.459	0.077	0.25	1	0	1
Sust_com	0.890	0.313	1	1	1	0	1
FSize	16.642	1.121	15.872	16.629	17.306	12.734	20.662
FPerf	0.086	0.083	0.042	0.071	0.122	−0.703	0.841
FLev	2.975	3.965	1.029	1.749	3.174	0.033	37.439
Env_industry	0.286	0.452	0	0	1	0	1
ESG_controv	85.388	24.993	82.350	100	100	1.79	100

is in line with other recent studies (Gull *et al.*, 2023a, 2023b, 2023c; Aboud *et al.*, 2024; Gull *et al.*, 2024) and which suggests the presence of brownwashing practices in the sample. A more detailed analysis shows that this variable presents a standard deviation of 0.162, indicating considerable variability in the alignment between environmental reporting and initiatives across firms. Regarding the main explanatory variables, the percentage of women on the board (Prop_wom) is 22%, with values ranging from 0 to 66.7%, highlighting notable disparity in board gender diversity across the sample, consistent with the Blau index (Blau), which has a mean of 0.33. In relation to the moderating factors, 29% of firms in our sample belong to an environmentally sensitive industry. The rating of ESG controversies is high, and averages around 85.39 out of 100 points, indicating that firms do not generally face substantial pressure concerning ESG misconduct, although there is a wide range from 1.79 to 100, thus reflecting varying degrees of media and stakeholder attention to ESG issues across firms.

Table 3 reports the bivariate correlations and the Variance Inflation Factor coefficients (VIF). In relation to the research hypothesis, it can be highlighted that board gender diversity is slightly correlated with Env_GAP. However, no clear effects are found for the moderating variables. Although the correlations cannot provide conclusive results – given that they only reflect bivariate associations – this table does allow us to rule out the existence of multicollinearity between independent variables included in the same econometric model, as correlations are less than 0.7 (Cooper and Schindler, 2003), and VIF values remain below 5 (Studenmund, 1997).

With regard to the first stage of our empirical analysis, Tables 4 and 5 respectively report the results from the GMM estimation model as well as all the additional endogeneity tests (fixed effect linear regression for panel data using the lagged independent variable, 2SLS and simultaneous equations). All the models suggest that board gender diversity leads to a reduction in environmental decoupling (Env_GAP) [8]. However, to gain a more accurate view of this effect, it is necessary to decompose the types of environmental gap.

Since these results might be insufficient to ascertain the particular effect of female directors on greenwashing and brownwashing environmental practices, in the second stage of our analysis the specific influence on each type of environmental decoupling is considered. Table 6 shows the results related to the influence of board gender diversity on the variables GREENWASH (columns 1 and 2) and BROWNWASH (columns 3 and 4). This

Table 3. Correlation matrix

Variables	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
(1) Env_GAP	-0.048**	-0.049**	-0.045**	0.063***	-0.172***	0.016	0.077***	-0.082***	0.081***	-0.104***	-0.019
(2) Prop_wom		0.958***	0.176***	0.062**	0.016	0.018	0.184***	-0.024	0.154***	-0.105***	-0.115***
(3) Blau			0.194***	0.092***	0.008	0.052**	0.205***	-0.033	0.158***	-0.088***	-0.133***
(4) BIndep				0.029	0.014	0.010	0.166***	-0.109***	0.103***	0.018	-0.020
(5) BSize					-0.134***	0.140***	0.368***	-0.072***	0.122	-0.035	-0.020
(6) BESG_Eexp						-0.023	-0.334***	0.207***	-0.158***	0.223***	0.231***
(7) Sust_com							0.070**	0.025	0.079***	-0.034	-0.005
(8) FSize								-0.522***	0.428***	-0.131***	-0.274***
(9) FPerf									-0.129***	0.070**	0.111
(10) FLev										-0.163***	-0.183***
(11) Env_industry											-0.101***
(12) ESC_controv											
VIF	1.10	1.12	1.07	1.22	1.24	1.03	2.22	1.54	1.28	1.13	1.18

Table 4. Effect of board gender diversity on environmental decoupling (GMM)

Variables	Env_GAP (1)	Env_GAP (2)
Prop_wom	-0.132* (0.071)	
Blau		-0.180* (0.092)
BIndep	-0.002 (0.047)	0.018 (0.054)
BSize	0.001 (0.002)	0.000 (0.002)
BESG_Eexp	-0.078*** (0.018)	-0.082*** (0.020)
Sust_com	0.006 (0.014)	0.008 (0.015)
FSize	0.016** (0.008)	0.021** (0.010)
FPerf	0.026 (0.041)	0.032 (0.045)
FLev	0.001** (0.001)	-0.002 (0.003)
Env_industry	0.137*** (0.050)	0.138** (0.058)
ESG_controv	0.002* (0.001)	0.002 (0.001)
Year effect	Yes	Yes
Observations	1,779	1,779
Wald chi ²	526.05***	450.61***
Arellano–Bond test AR(2)	0.90	0.69
AR(2) <i>p</i> -value	0.367	0.493
Sargan test	28.08	30.50
Sargan <i>p</i> -value	0.514	0.340

Note(s): Standard errors in parentheses. ****p* < 0.01, ***p* < 0.05, **p* < 0.1

table highlights that women in the boardroom increase greenwashing and reduce brownwashing.

These findings suggest that female directors generally shift the balance between environmental reporting and initiatives, and that the greater emphasis on reporting implied by our theoretical arguments may have different effects on environmental decoupling. On the one hand, the positive effect of board gender diversity on GREENWASH implies that female directors intensify greenwashing in companies that are overreporting. On the other hand, the negative effect of board gender diversity on BROWNWASH implies that female directors reduce brownwashing in companies that are underreporting. Hence, our *H1* can be supported.

Considering each type of decoupling reveals more nuanced results. In line with our theoretical expectations, while female personal traits lead to a stronger orientation toward environmental decisions, women in the boardroom – albeit not necessarily intentionally – may influence environmental decoupling by exerting a more immediate and effective impact on environmental reporting than on initiatives. This can occur given the potentially limited influence of female directors, particularly regarding environmental initiatives, which involve investment and which are characterized by high uncertainty. As a result, our findings point to a complex dynamic; while greenwashing appears to grow – albeit not necessarily due to unethical reasons – brownwashing is mitigated.

Furthermore, the DiD results shown in Table 7 and Figure 2 reveal a negative and statistically significant interaction term, suggesting that the introduction of the California quota law is associated with a reduction in environmental decoupling without considering the environmental gap sign. Furthermore, breaking down decoupling according to its sign offers more conclusive insights; female directors increase greenwashing whereas they decrease brownwashing with the introduction of the California quota law.

Table 5. Endogeneity tests

Variables	(1)		(2)		(3)	
	FE	2SLS	First stage	Second stage	Simultaneous equations	
L.Prop_wom	-0.073* (0.037)					
Prop_wom						
L.Blau	-0.075** (0.032)	-0.413* (0.248)			-0.084** (0.037)	
Blau						-0.081** (0.035)
Prop_F_nomin			0.026* (0.014)			
BTenure			-0.003*** (0.001)			
BIndep	-0.006 (0.058)	0.052 (0.078)	0.227*** (0.039)	0.110 (0.098)	-0.080* (0.044)	-0.077* (0.044)
BSize	0.001 (0.002)	-0.001 (0.003)	-0.001 (0.002)	-0.001 (0.003)	0.004** (0.002)	0.004** (0.002)
BESG_exp	0.021 (0.058)	0.173** (0.074)	0.012 (0.044)	0.211*** (0.077)	-0.050*** (0.009)	-0.050*** (0.009)
Sust_com	0.017 (0.021)	0.061** (0.024)	0.004 (0.015)	0.063** (0.025)	0.002 (0.011)	0.003 (0.011)
FSize	0.005 (0.009)	0.019* (0.011)	0.002 (0.007)	0.021* (0.012)	-0.006 (0.005)	-0.006 (0.005)
FPerf	-0.116** (0.046)	-0.089 (0.056)	0.088*** (0.033)	-0.068 (0.062)	-0.127** (0.052)	-0.127** (0.052)
FLev	0.002 (0.001)	0.002 (0.001)	-0.001 (0.001)	0.002 (0.001)	0.003*** (0.001)	0.003*** (0.001)
Env_industry					-0.023*** (0.008)	-0.023*** (0.008)
ESG_controv					0.000 (0.000)	0.000 (0.000)
Year effect	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,002	1,534	1,724	1,534	2,002	2,002
R-squared	0.008	0.034	0.455	-0.041	0.046	0.046
F-test	1.78*	4.26***	67.31***	4.00***		
Sargan test		1.318		0.506		
Sargan test p-value		0.251		0.477		

Note(s): Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 6. Analysis by GAP sign

Variables	GREENWASH (1)	GREENWASH (2)	BROWNWASH (3)	BROWNWASH (4)
Prop_wom	0.526*** (0.169)			
Blau		0.489*** (0.166)		−0.201** (0.088)
Blndep	−0.197** (0.098)	−0.057 (0.078)	−0.127* (0.065)	0.142*** (0.049)
BSize	−0.011** (0.005)	0.002 (0.004)	0.104*** (0.037)	0.003 (0.003)
BESG_exp	0.038 (0.027)	−0.017 (0.018)	0.001 (0.001)	−0.055** (0.022)
Sust_com	0.006 (0.023)	−0.016 (0.014)	0.004 (0.009)	0.008 (0.013)
FSize	0.109*** (0.038)	0.011 (0.016)	−0.004 (0.005)	−0.011 (0.020)
FPerf	0.519*** (0.171)	0.469* (0.263)	0.001 (0.027)	−0.093 (0.086)
FLev	−0.006 (0.006)	−0.000 (0.002)	−0.001 (0.001)	−0.008*** (0.003)
Env_industry	0.026 (0.031)	0.026 (0.029)	−0.063 (0.049)	−0.010 (0.011)
ESG_controv	0.000 (0.001)	0.001 (0.001)	−0.000 (0.001)	0.001 (0.001)
Year effect	Yes	Yes	Yes	Yes
Observations	492	492	1,287	1,287
Wald χ^2	290.70***	469.26***	3599.12 ***	2596.15***
Arellano–Bond test AR(2)	0.91	1.29	0.39	−0.09
AR(2) <i>p</i> -value	0.363	0.196	0.698	0.932
Sargan test	34.47	26.74	46.28	41.40
Sargan <i>p</i> -value	0.186	0.180	0.117	0.211

Note(s): Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 7. Difference-in-differences (DiD) results

Variables	Env_GAP (1)	Env_GAP (2)	GREENWASH (3)	GREENWASH (4)	BROWNWASH (5)	BROWNWASH (6)
Prop_wom	-0.212* (0.124)		0.483 (0.543)		-0.036 (0.094)	
Blau		-0.144 (0.119)		-0.099 (0.449)		-0.060 (0.123)
California_dummy	0.255** (0.102)	0.248** (0.100)	-1.165* (0.676)	-0.079 (0.502)	0.838* (0.432)	0.883** (0.450)
Year 2018	0.038** (0.017)	0.035** (0.017)	-0.282** (0.137)	-0.094* (0.057)	0.012 (0.013)	0.008 (0.014)
California_dummy*Year 2018	-0.339* (0.204)	-0.342*	1.655** (0.839)	0.655*	-0.818* (0.451)	-0.740*
Blndep	0.014 (0.058)	0.014 (0.059)	-0.817** (0.381)	-0.790* (0.415)	0.322*** (0.104)	0.339*** (0.117)
BSize	0.004 (0.003)	0.004 (0.003)	-0.080** (0.035)	-0.059*** (0.016)		
BESG_exp	-0.071*** (0.026)	-0.070*** (0.025)	-0.484 (0.296)	-0.331 (0.214)	-0.041* (0.021)	-0.006 (0.009)
Sust_com	0.001 (0.020)	-0.001 (0.018)	0.291 (0.539)	-0.144 (0.114)	-0.010 (0.112)	0.065 (0.108)
FSize	0.012 (0.010)	0.010 (0.009)	0.289** (0.125)	0.342*** (0.114)	-0.020 (0.026)	-0.017 (0.023)
FPerf	-0.001 (0.054)	-0.002 (0.053)	0.852* (0.448)	0.436 (0.393)	-0.100 (0.098)	-0.060 (0.091)
FLev	0.001 (0.001)	0.001 (0.001)	0.017 (0.019)	0.005 (0.015)	-0.006*** (0.002)	-0.004* (0.002)
Env_industry	0.103 (0.066)	0.094 (0.063)	0.011 (0.294)	-0.067 (0.203)	0.002 (0.014)	0.007 (0.015)
ESG_controv	0.001 (0.001)	0.001 (0.001)	0.006* (0.003)	0.005 (0.003)	-0.000 (0.000)	-0.000 (0.000)
Observations	1,779	1,779	492	492	1,287	1,287
Wald chi ²	363.56***	417.67	86.31***	70.78***	2,404.79***	1,645.58***
Arellano–Bond test AR(2)	-0.22	-0.01	1.34	1.30	-0.73	-0.40
AR(2) p-value	0.827	0.992	0.180	0.194	0.466	0.690
Sargan test	25.12	29.11	9.04	13.15	49.15	45.23
Sargan p-value	0.622	0.407	0.828	0.590	0.209	0.339

Note(s): Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

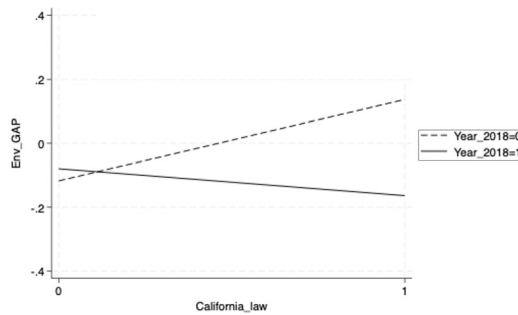


Figure 2. Marginal effect plots for the interaction between board gender diversity and the California quota law

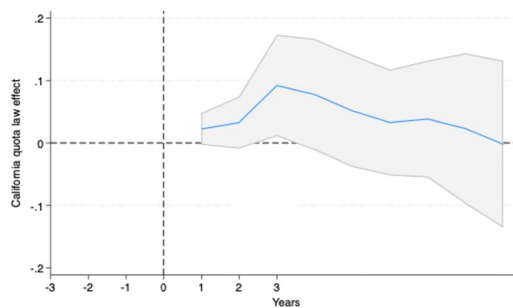


Figure 3. Pre-trend analysis of the California quota law

The event study plot (Figure 3) confirms the validity of the parallel trend assumption, as all pretreatment coefficients ($t < 0$) are statistically indistinguishable from zero, thus demonstrating that trends were not diverging before intervention.

As for the third stage of our empirical study, moderation analyses are performed to examine how external pressures driven by the industry and by the level of a firm's ESG controversies can moderate the influence of board gender diversity on environmental decoupling.

Table 8 shows that environmentally sensitive industries significantly moderate the effect of female board members on environmental decoupling. While female directors appear to reduce overall Env_GAP (columns 1 and 2), a disaggregated (columns 3–6) analysis reveals that board gender diversity leads to an increase in greenwashing and a decrease in brownwashing. This pattern is consistent with our theoretical arguments due to heightened societal and regulatory pressures in these sectors. In addition, Figures 4–6 present the marginal effect plots for the interaction between board gender diversity and environmental industry and its effect on the overall environmental gap –greenwashing and brownwashing – respectively. Hence, $H2$ is supported.

Table 9 fails to show any significant moderating effect of firm-level ESG controversies on the influence of board gender diversity on overall environmental decoupling (columns 1 and

Table 8. Moderation analysis by environmentally sensitive industries

Variables	Env_GAP (1)	Env_GAP (2)	GREENWASH (3)	GREENWASH (4)	BROWNWASH (5)	BROWNWASH (6)
Prop_wom*Env_industry	-1.342* (0.792)		3.153*** (1.002)		-0.833* (0.481)	
Prop_wom	0.262 (0.217)		-0.028 (0.294)		0.142 (0.153)	
Blau*Env_industry						
Blau		-0.965* (0.538)		1.713** (0.804)		-1.672** (0.849)
Blndep		0.198 (0.167)		0.042 (0.288)		0.194 (0.281)
BSize	-0.080 (0.078)	-0.049 (0.071)	-0.605 (0.750)	-0.202 (0.698)	0.051 (0.037)	0.023 (0.067)
BESG_exp	0.005* (0.003)	0.003 (0.002)	-0.029*** (0.011)	-0.014 (0.009)	0.002 (0.002)	0.003 (0.002)
Sust_com	-0.019 (0.077)	-0.061 (0.067)	-0.009 (0.060)	-0.012 (0.061)	-0.012 (0.009)	0.088 (0.069)
FSize	0.010 (0.015)	0.007 (0.016)	-0.024 (0.017)	-0.029 (0.021)	0.002 (0.008)	-0.005 (0.016)
FPerf	0.009 (0.009)	0.006 (0.009)	0.167*** (0.048)	0.091** (0.041)	-0.004 (0.004)	0.011 (0.013)
FLev	-0.028 (0.052)	-0.025 (0.056)	0.621*** (0.191)	0.700*** (0.218)	-0.010 (0.030)	-0.047 (0.055)
Env_industry	0.001 (0.001)	0.001 (0.001)	-0.008** (0.004)	-0.003 (0.003)	-0.001 (0.000)	-0.000 (0.000)
ESG_controv	0.395** (0.173)	0.424** (0.196)	-0.667*** (0.219)	-0.569** (0.259)	0.185* (0.104)	0.525* (0.269)
Year effect	0.000 (0.000)	0.000 (0.000)	0.002** (0.001)	0.001 (0.001)	-0.000 (0.001)	0.000 (0.001)
Observations	Yes	Yes	Yes	Yes	Yes	Yes
Wald chi ²	1,779	1,779	492	492	1,287	1,287
Arellano-Bond test AR(2)	435.37***	334.94***	354.29***	358.57***	5776.93***	2408.52***
AR(2) p-value	0.55	0.34	0.80	1.05	0.14	-0.48
Sargan test	0.580	0.733	0.425	0.295	0.890	0.635
Sargan p-value	21.92	24.82	33.29	37.69	42.81	26.99
	0.345	0.208	0.154	0.304	0.202	0.136

Note(s): Standard errors in parentheses. ***p <0.01, **p <0.05, *p <0.1

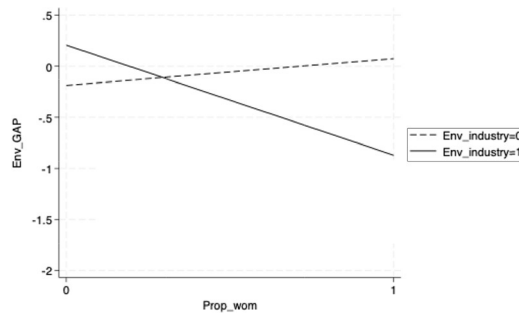


Figure 4. Marginal effect plots for the interaction between board gender diversity and environmental industry and its effect on the overall environmental gap

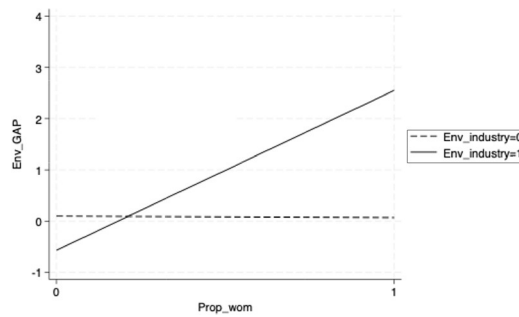


Figure 5. Marginal effect plots for the interaction between board gender diversity and environmental industry and its effect on GREENWASH

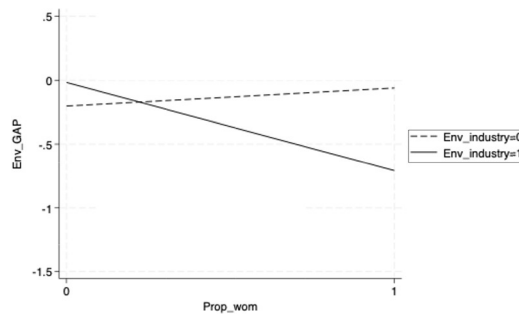


Figure 6. Marginal effect plots for the interaction between board gender diversity and environmental industry and its effect on BROWNWASH

Table 9. Moderation analysis by firm-level of ESG controversies^a

Variables	Env_GAP (1)	Env_GAP (2)	GREENWASH (3)	GREENWASH (4)	BROWNWASH (5)	BROWNWASH (6)
Prop_wom*ESG_controv	-0.012 (0.009)		0.033*** (0.010)		-0.003 (0.007)	
Prop_wom	0.942 (0.767)		-2.349*** (0.900)		0.258 (0.655)	
Blau*ESG_controv						
Blau		-0.001 (0.013)		0.097** (0.039)		0.012 (0.012)
BIndep	0.013 (0.047)	-0.073 (1.186)	-0.147* (0.086)	-7.663** (3.487)		-1.137 (1.166)
BSize	0.003* (0.002)	0.013 (0.064)	0.003 (0.004)	-0.247 (0.228)	0.102** (0.042)	0.172*** (0.054)
BESG_exp	-0.007 (0.050)	0.000 (0.002)	0.113* (0.062)	-0.003 (0.009)	0.000 (0.001)	-0.001 (0.002)
Sust_com	-0.002 (0.014)	-0.102 (0.063)	0.022 (0.019)	0.218** (0.088)	-0.061 (0.042)	-0.048 (0.048)
FSize	0.005 (0.007)	0.003 (0.018)	0.022 (0.011)	0.078 (0.072)	0.004 (0.009)	-0.000 (0.013)
FPerf	-0.016 (0.038)	0.018 (0.012)	0.005 (0.011)	0.062 (0.039)	-0.001 (0.005)	0.019 (0.012)
FLev	0.001 (0.000)	0.034 (0.047)	-0.129 (0.109)	0.328 (0.352)	0.033 (0.031)	0.613*** (0.210)
Env_industry	-0.003 (0.016)	0.002** (0.001)	0.000 (0.001)	-0.009 (0.010)	-0.000 (0.000)	-0.001 (0.000)
ESG_controv	0.003 (0.002)	0.144** (0.067)	-0.024 (0.032)	0.093 (0.145)	0.020* (0.011)	0.012 (0.013)
Year effect	Yes	0.003 (0.005)	-0.007*** (0.002)	-0.033** (0.013)	0.002 (0.002)	-0.004 (0.005)
Observations	1,779	Yes	Yes	Yes	Yes	Yes
Wald chi ²	1,779	492	492	492	1,287	492
Arellano-Bond test AR(2)	735.37***	371.63***	454.95***	144.46***	6125.69***	414.86***
AR(2) p-value	1.26	0.61	1.25	1.00	0.48	1.04
Sargan test	0.207	0.541	0.210	0.315	0.629	0.297
Sargan p-value	32.10	30.06	34.54	21.81	47.18	26.85
	0.316	0.311	0.184	0.241	0.235	0.140

Notes(s): Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$; ^aWe performed a sensitivity check by comparing the means of the dependent variable across sub-samples. Although statistically significant due to the large sample size, the absolute difference in disclosure levels is less than 2%, confirming the representativeness of the controversy sub-sample

2). Nevertheless, the specific analysis of each type of environmental gap highlights that female directors do have a greater positive effect on greenwashing (columns 3 and 4) when firm-level ESG controversies remain low. Contrary to our expectations, the influence of board gender diversity on environmental decoupling appears to be stronger in firms with lower ESG controversies, which differs from the pattern observed with environmentally sensitive industries. This divergence may be attributed to the difference in the level of analysis; while environmentally sensitive industries capture sector-wide pressures that broadly shape corporate behaviors, ESG controversies represent firm-specific reputational events that impose immediate constraints or risks. Therefore, our *H3* cannot be accepted. From a theoretical perspective, the firm-specific nature of ESG controversies may lead to more cautious or strategic behaviors and limit any widespread influence on environmental gap under higher firm-specific pressures. What is more, [Figure 7](#) represents the marginal effect plots for the interaction between board gender diversity and ESG controversies and its effect on greenwashing. This reflects a more complex interplay of pressures and governance dynamics than initially hypothesized.

Finally, as regards the fourth stage of our empirical analysis, which consists of the triple interaction between *Prop_wom*, *Env_industry* and *ESG_controv*, [Table 10](#) confirms that the effect of board gender diversity on environmental decoupling is simultaneously moderated by pressures from the industry and pressures from the firm itself. Specifically, results show that female directors lead to increased greenwashing in firms operating in environmentally sensitive industries and that exhibit low levels of ESG controversies (columns 1 and 2). This pattern suggests that female directors respond strongly to sector-wide external pressures by enhancing reporting when companies overreporting results in increased environmental decoupling, particularly when firm-specific reputational risks are low. These findings suggest that women on boards may be more proactive in environmental decoupling under conditions of high industry pressure but limited firm-level controversies. When firm-level ESG controversies are high, the strategic constraints and reputational risks appear to temper this effect. In addition, [Figure 8](#) exhibits marginal effect plots for the triple interaction between *Prop_wom*, *Env_industry* and *ESG_controv* and its effect on GREENWASH.

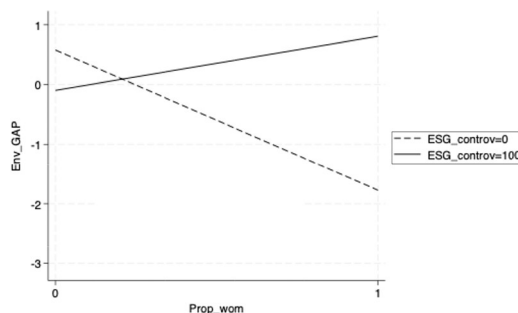


Figure 7. Marginal effect plots for the interaction between board gender diversity and ESG controversies and its effect on GREENWASH

Table 10. Triple interaction between board gender diversity, environmentally sensitive industries and ESG controversies

Variables	GREENWASH (1)	GREENWASH (2)	BROWNWASH (3)	BROWNWASH (4)
Env_industry*Prop_wom*ESG_controv	0.028*** (0.009)		0.000 (0.010)	0.013 (0.010)
Prop_wom	-1.820* (0.953)	0.069*** (0.022)	-0.262 (0.823)	-1.240 (0.899)
Env_industry*Blau*ESG_controv		-5.275*** (2.089)		0.167*** (0.053)
Blau		-0.286* (0.163)	0.103** (0.046)	-0.002 (0.002)
BIndep	-0.105 (0.091)	-0.002 (0.006)	0.001 (0.001)	-0.022 (0.044)
BSize	0.005 (0.003)	0.136* (0.075)	-0.038 (0.042)	0.004 (0.015)
BESG_exp	0.131** (0.056)	0.030 (0.055)	0.007 (0.010)	0.015 (0.011)
Sust_com	0.022 (0.025)	-0.000 (0.016)	-0.004 (0.006)	0.487*** (0.164)
FSize	-0.008 (0.008)	-0.395** (0.189)	0.013 (0.033)	-0.001* (0.001)
FPerf	-0.127 (0.108)	0.000 (0.002)	0.059 (0.061)	-0.106 (0.136)
FLev	0.000 (0.001)	0.053 (0.232)	-0.000 (0.002)	-0.004 (0.004)
Env_industry	-0.072 (0.112)	-0.023*** (0.008)	1.287	1.287
ESG_controv	-0.007*** (0.002)	492	0.36	0.00
Observations	492	1.17	0.36	0.996
Arellano–Bond test AR(2)	1.28	0.241	0.722	42.12
AR(2) <i>p-value</i>	0.199	36.74	42.07	0.191
Sargan test	33.70	0.343		
Sargan <i>p-value</i>	0.211			

Note(s): Standard errors in parentheses. ****p* < 0.01, ***p* < 0.05, **p* < 0.1

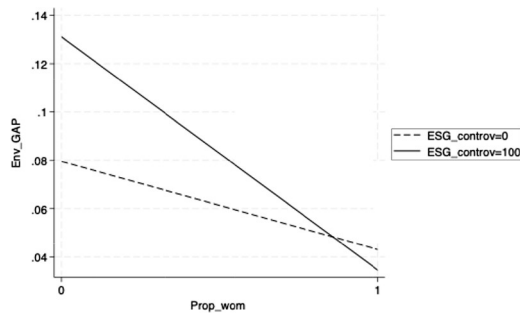


Figure 8. Marginal effect plots for the triple interaction between Prop_wom, Env_industry and ESG contro and its effect on GREENWASH

5. Discussion and conclusion

This study analyses the effect of board gender diversity on environmental decoupling and provides new insights in this research area. Our evidence underscores that female directors influence environmental decoupling, although their impact varies depending on whether this decoupling leads to greenwashing or brownwashing environmental practices. Our results also indicate that the effect of board gender diversity on environmental decoupling is contingent upon both general industry pressures and firm-level pressures arising from ESG controversies. Furthermore, how female directors respond to these pressures differs significantly.

Our paper extends previous research in several ways. First, in line with recent literature (Gull *et al.*, 2024; Palea *et al.*, 2025), we emphasize the importance of considering both dimensions of ESG decoupling, which determines corporate transparency and ultimately contributes to firms' sustainable development: greenwashing and brownwashing. Firms may engage in greenwashing practices to positively influence stakeholder perceptions and to enhance profitability (Kim and Lyon, 2015) as well as align with social and regulatory trends (Delmas and Burbano, 2011). Conversely, companies might resort to brownwashing strategies to mitigate dissent from investors who do not align with ESG investments and to avoid pressure from stakeholders who may label the firm unfavorably (Montgomery *et al.*, 2024). Since our findings show different board responses to greenwashing and brownwashing practices, our paper calls for further research that considers the specific direction of this decoupling.

Second, our results strengthen academic debates concerning the role of the board in how firms contribute to sustainable development through their impact on environmental strategies (Velte, 2023; Khatib and Sulimany, 2025). In addition to the external factors, such as pressures from regulators and stakeholders, it is crucial to understand how internal corporate governance mechanisms may shape environmental decoupling (Shahab *et al.*, 2022; Zhao *et al.*, 2022) so that effective measures may be adopted. Our findings indicate that boards serve as a critical conduit to determine both environmental greenwashing and brownwashing practices.

Third, we extend recent discussions regarding how female directors deal with environmental decoupling. In this regard, we reconcile recent contradictory evidence (Eliwa *et al.*, 2023; Gull *et al.*, 2023b; Shahab *et al.*, 2022) by showing that the effect of board gender diversity on environmental decoupling may be negative or positive depending on the sign of this decoupling. From a gender socialization perspective (Aryssi *et al.*, 2019;

Harjoto and Rossi, 2019; Ramon-Llorens *et al.*, 2021), our findings support that female directors may have a stronger orientation toward environmental issues. However, the increase in environmental reporting does not seem to be aligned with enhanced environmental initiatives. According to our theoretical arguments, female directors may shift the balance toward environmental reporting over internal initiatives because reporting is easier to implement, less resource-intensive and more accessible for minority directors. Theoretically, this suggests that while female directors strive to discharge the firm's accountability through enhanced transparency, they may face structural barriers when attempting to implement substantive environmental changes. In firms with higher levels of environmental actions, female directors thus reduce decoupling by mitigating brownwashing practices, whereas in firms with lower levels of environmental actions, female directors trigger an increase in decoupling by incurring in greenwashing practices. These findings call into question the role of female directors in environmental decoupling. While our results may not reflect intentionally unethical behavior, our study opens the door to new theoretical frameworks that could explain this situation. In this regard, the influence of female directors may be observed initially in reporting issues, which are easier to implement (Matuszak *et al.*, 2019; Liang *et al.*, 2022). Likewise, it is possible that women may not exert enough influence to impact complex environmental actions such as the implementation of environmental policies (Cook and Glass, 2015).

Fourth, our findings also align with recent research which suggests that contextual approaches are essential for a more nuanced understanding of female director influence on ESG decision-making (Endrikat *et al.*, 2021; Ramon-Llorens *et al.*, 2021; García-Meca and Martínez-Ferrero, 2025). In this vein, recent papers have posited that the effect of female directors on ESG decoupling may be contingent on the level of religiosity (Eliwa *et al.*, 2023) and economic policy uncertainty (Qureshi *et al.*, 2024b). However, previous studies have suggested that external pressures are a key driver of environmental decoupling (Cho *et al.*, 2015; Kim *et al.*, 2017; Tashman *et al.*, 2019). Our paper adds new evidence by highlighting that these external pressures also moderate board influence on environmental decoupling. In particular, the role of female directors in environmental decoupling varies depending on broader industry pressures as well as firm-specific pressures related to the level of ESG controversies faced by the organization. Our results suggest that different types of pressures (sector and firm-specific) operate differently, which is a key nuance that highlights the complexity of governance dynamics. This indicates that female directors act as key actors who respond to stakeholder demands for transparency to ensure corporate accountability, although the effectiveness of this response depends on the specific firm-level and industry context. In this vein, these findings emphasize the need for future studies to consider various types of pressures when examining the influence of boards and, more specifically, how gender diversity might impact environmental strategies.

Given the importance of both board gender diversity and ESG decoupling –and particularly its environmental dimension – on society and sustainable development, our paper has direct implications for firms, regulators, professionals, society and researchers alike. First, our findings offer valuable insights for firms by providing useful evidence to strengthen the integrity of their accountability mechanisms. Shareholders, in particular, should be vigilant when assessing boards in industries that are sensitive to environmental concerns. Our study also has managerial implications, since companies must recognize that enhancing environmental reporting without aligning it to actual environmental actions may lead to environmental decoupling. This poses both ethical risks, such as misleading stakeholders, and economic risks, including possible regulatory penalties, thus compromising substantive sustainable development. In this context, companies should

consider redesigning internal ESG dashboards and executive incentive schemes to reward not only reporting but also the implementation of verifiable environmental initiatives. To monitor this, firms can implement decoupling KPIs that measure the variance between ESG reporting scores and internal performance metrics, thereby allowing boards to set thresholds when reporting differs from operational progress. In addition, female directors should be included in strategic committees beyond CSR or compliance to enhance their influence on environmental action and not just on reporting.

Second, international professional organizations and policymakers have issued recommendations regarding board gender diversity. Our findings offer a significant business case for both refining existing legislation and for providing practical recommendations for regulators and industry practitioners. It is particularly crucial to recognize that the decision-making behavior of female directors concerning environmental matters may be heavily influenced by external pressures – both from the industry and the firm itself. In addition, our evidence provides valuable insights for regulatory bodies by stressing the importance of these external pressures in the process of environmental decoupling. This issue is likely to present a substantial challenge for auditors, who must remain particularly vigilant on the matter. To address this challenge, ESG assurance providers and auditors should monitor for overreporting practices to ensure that corporate transparency is substantive rather than purely symbolic. Sustainability officers could implement internal checks to detect early gaps between reporting and action.

Recent regulatory developments – particularly the EU Corporate Sustainability Reporting Directive (CSRD) – underscore the importance of aligning reported environmental information with actual performance, thus emphasizing dual materiality. Our findings highlight the need to assess firms' environmental decoupling practices. Decoupling indicators can guide auditors and assurance providers in determining the appropriate scope and depth of verification – whether limited or reasonable – by focusing on areas where reporting may overstate or understate actual environmental performance.

Third, environmental decoupling can have significant societal consequences, since it hinders the firm's contribution to sustainable development goals. Excessive disclosure without the corresponding action (greenwashing) can erode stakeholder trust and misallocate capital, whereas under-disclosure (brownwashing) may conceal environmental harm, hinder stakeholder recognition of genuine environmental efforts and increase the risk of regulatory noncompliance. Boards play a crucial supervisory role in mitigating these two risks by implementing internal oversight mechanisms, aligning executive incentives with verified environmental performance, and by ensuring that reporting practices accurately reflect the firm's actual environmental actions.

Fourth, from an academic perspective, our study highlights the need to develop new theoretical frameworks so as to gain a deeper understanding of the role of board gender diversity. In this regard, our findings reinforce the notion that female directors may influence firms' sustainable development by affecting both environmental brownwashing and greenwashing practices (Kim and Lyon, 2015; Gull *et al.*, 2023c). It is important to expand theoretical research into the reasons why women in the boardroom may increase greenwashing practices. Furthermore, our research adds to the ongoing academic discourse by underscoring the importance of considering the contextual factors within which directors make their decisions, particularly a firm's external pressures. Finally, we contribute to the power versus presence debate by suggesting that while female directors may influence reporting, their capacity to shape internal environmental strategies may remain constrained.

This study acknowledges several limitations that should be considered when conducting future research. First, although the USA provides a relevant scenario for examining corporate governance mechanisms and ESG decoupling, future studies could explore different legal and institutional environments, as the influence of boards may vary across contexts. In addition, our sample focuses on large publicly listed firms, such that the relatively small variations in firm size within our sample may also be viewed as a limitation. Future research could therefore expand to include small and medium-sized enterprises. Another promising avenue for investigation could involve exploring the specific characteristics of female directors and their impact on environmental decoupling. In particular, it would prove valuable to delve deeper into the actual power of female directors –specifically, their influence within key board committees– rather than focusing solely on their numerical representation on boards. Such an analysis would provide a more comprehensive understanding of how gender can shape board dynamics. Furthermore, it is important to note that LSEG ESG scores do not cover all companies, which may affect the generalizability of our findings, as the data set tends to overrepresent larger, publicly traded firms with greater disclosure requirements and stakeholder pressure. As a result, the degree of environmental decoupling and the observed effects of board gender diversity may differ from those in smaller or less regulated firms. Finally, it would be interesting to incorporate text mining techniques for environmental reporting in future studies, as these methods might offer valuable alternatives to commercial databases.

Notes

- [1.] Also referred to as the Corporate Sustainability Reporting Directive (CSRD).
- [2.] Commonly referred to as the Green Claims Directive.
- [3.] Gender Equality Strategy 2020-2025; US National Strategy on Gender Equality.
- [4.] Additional analyses were conducted using two- and three-year windows between internal and external actions. Results were consistent and are available upon request.
- [5.] To assess sensitivity to normalization, we conducted a robustness check using the natural logarithm of the variable ENV_GAP. Results remained consistent with the main analysis, confirming that our findings are not driven by the use of min-max normalization.
- [6.] ESG controversies is obtained from the LSEG Workspace database, the world's largest ESG rating database and one that is used in numerous studies (Aouadi and Marsat, 2018; Gallego-Álvarez and Pucheta-Martínez, 2021, Agnese *et al.*, 2023). The LSEG Workspace database establishes that this variable measures a company's exposure to environmental, social, and governance controversies and negative events reflected in global media. The default value of all controversy measures is 0, whereas companies with no controversies obtain a score of 100 (LSEG Data and Analytics, 2024).
- [7.] We identify two kinds of sectors to reflect a meaningful contrast between firms that are generally more exposed to ESG-related pressures and expectations, and those that are typically less exposed. This classification allows us to investigate the role of ESG sensitivity more precisely, which is central to our research question. The SIC codes for industries classified as environmentally sensitive are those between 1,000 and 1,500; those between 2,800 and 3,000; and those between 4,900 and 5,000.
- [8.] To address potential concerns regarding sample selection bias and data exclusions, we conducted an additional robustness check using the full sample, retaining all outliers. The results remained qualitatively and statistically consistent with our main analysis, confirming that our findings are not driven by the exclusion of these observations.

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Appendix

To construct our dependent variable, we rely on the description of items made by [Gallego-Álvarez and Pucheta-Martínez \(2021\)](#) related to CSR reporting items, provided by the LSEG Eikon database. In particular, we focus solely on items related to the environmental pillar and go a step further by classifying them between initiative and disclosure items, taking into account the description of each variable in the LSEG Workspace database. Each of these items is shown in [Table A1](#) and is measured as a dummy variable, taking 1 if firms disclose/have such a policy, and 0 otherwise. Considering the available data, our dependent variable is the difference between 22 items related to reporting, measured one year ahead, and 21 items related to initiatives, both normalized considering the maximum and minimum values of the sample.

Table A1. Environmental reporting/initiative items

Items	Classification
Resource reduction policy	Initiative
Water efficiency policy	Initiative
Energy efficiency policy	Initiative
Sustainable packaging policy	Initiative
Environmental supply chain policy	Initiative
Environment management team	Initiative
Environment management training	Initiative
Environmental materials sourcing	Disclosure
Toxic chemicals reduction	Disclosure
Renewable energy use	Initiative
Green buildings	Disclosure
Environmental supply chain management	Initiative
Environmental supply chain partnership termination	Disclosure
Land environmental impact reduction	Disclosure
Emissions reduction policy	Initiative
Emissions reduction targets	Initiative
Biodiversity impact reduction	Disclosure
Emission trading	Disclosure
NOx and SOx emissions reduction	Disclosure
VOC emissions reduction	Disclosure
Particulate matter emissions reduction	Disclosure
e-waste reduction	Disclosure
Total waste reduction	Disclosure
Environmental restoration initiatives	Disclosure
Staff transportation impact reduction	Disclosure
Eco-design products	Disclosure
Environmental products	Disclosure
Noise reduction	Initiative
Hybrid vehicles	Initiative
Environmental asset under management	Disclosure
Environmental project financing	Disclosure
Nuclear	Initiative
Labeled wood	Disclosure
Organic products initiatives	Disclosure
Take-back and recycling	Disclosure
Responsible environmental product use	Disclosure
(continued)	

Table A1. Continued

Items	Classification
GMO products	Initiative
Animal testing	Initiative
Animal testing cosmetics	Initiative
Animal testing reduction	Initiative
Renewable clean energy products	Initiative
Water technologies	Initiative
Sustainable building products	Initiative

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